Technological Forecasting & Social Change 197 (2023) 122881

Contents lists available at ScienceDirect

Technological Forecasting & Social Change

journal homepage: www.elsevier.com/locate/techfore

The impact of the artificial intelligence industry on the number and structure of employments in the digital economy environment

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ARTICLE INFO

Keywords:
The digital economy
AI industry
Employment structure
Labor
AI

ABSTRACT

This study endeavors to examine the intricate impact of the industries related to artificial intelligence (AI) on both the quantity of labor employment in terms of quantity and the evolving structure of the job market. The study constructs a mechanism and theoretical model to elucidate labor employment dynamics and structural transformations within the AI industry. The research findings reveal discernible trend in China's labor force characterized by a heightened emphasis on education and increased employment opportunities. This study makes a substantial contribution to the field by constructing innovative mechanisms and theoretical frameworks that facilitate a profound comprehension of the intricacies surrounding labor employment and structural changes in the AI industry. This comprehension is grounded in an analysis of the prevailing employment quantity and structure. This study is underpinned by an examination of employment distribution across diverse industries and a meticulous analysis of the workforce's skill diversity. By thoroughly analyzing China's digital economy policies and the strides made in digital technologies, the study formulates a set of recommendations aimed at fostering national economic growth and achieving successful digital transformation. The primary innovation of this study resides in its exploration of the impact of the AI industry with respect to China's labor employment structure through the lens of a Marxist theoretical perspective. Consequently, the study extends both innovative significance and practical value.

1. Introduction

The rapid progression of science and technology has precipitated the widespread integration of core technologies, notably artificial intelligence (AI), big data, and blockchain, across diverse sectors of society. In China, AI-related industries have experienced notably swift expansion. Noteworthy examples encompass mobile payments, virtual online shopping, live e-commerce deliveries, online education, and AI-powered educational materials. These industries have spearheaded the intelligent evolution of their respective sectors. However, alongside their contributions to national economic advancement, it is imperative to acknowledge the potential substitution effect on labor employment (Zhou et al., 2020; Tschang and Almirall, 2021). For example, the implementation of AI technologies in intelligent factory production lines has resulted in the automation of approximately 50 % of tasks previously undertaken by human labor. Similarly, the pervasive adoption of intelligent teller machines within the financial sector has led to a substantial decrease in employment for traditional tellers and customer service personnel. Consequently, the incorporation of AI across various industries has influenced employment rates and has ushered in substantial transformations in lifestyle, economic frameworks, and social dynamics (Kaiming, 2021; Chen et al., 2022; Aghion et al., 2019). Therefore, given the expeditious growth of AI-related industries, comprehending their impact on employment quantity and structure has emerged as a focal point for scholars in cognate fields.

Employment quantity and structure stand as pivotal indicators of a nation's or region's development status. In the context of the digital economy, the substitution of human employment with AI technology across various domains has ignited considerable discourse. The widespread integration of AI technology has indisputably emerged as an undeniable trend, inevitably impacting human employment and, consequently, the configuration of the labor force. As AI technology continues its pervasive application and promotion in pertinent fields, it has engendered alterations in the number of employed individuals, the distribution of the workforce among industries and occupations, and the evolving societal demands for skills (Frank et al., 2019). However, despite the profound integration of AI technology into virtually every facet of life, consensus regarding its impact on employment structure

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https://doi.org/10.1016/j.techfore.2023.122881
Received 10 November 2022; Received in revised form 20 September 2023; Accepted 20 September 2023
Available online 11 October 2023
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Furthermore, an examination of the employment structure of the labor force is conducted. The employment structure typically pertains to allocating the working population across different industries within a country or region's social system (Lakshmi and Bahli, 2020). Fig. 2 presents the categorization of labor employment industries within the social system.

In Fig. 2, the labor force is predominantly engaged in various industries within the social system, including transportation, culture, tourism, government affairs, finance, education, social networking, environmental protection, water, and electricity. The employment structure of the labor force encompasses several dimensions, such as industry, skill, occupation, age, and gender, providing insights into the workforce composition in a given country or region during a specific period. Among these dimensions, industry classification is crucial in examining the labor force's structure. Furthermore, the skill structure of the labor force serves as a significant measure for assessing the quality of a country's workforce (Chalmers et al., 2021). Consequently, this study primarily investigates the distribution of employees across different industries and the proportion of employees with varying skill levels.

3.2. *Analysis of the impact of AI-related industrial structure changes and labor employment*

The current state of China's AI industry exhibits a comprehensive industrial structure centered around AI technology. This structure can be analyzed at three levels: the basic, technology, and application layers. Each level possesses its distinct technical and industrial characteristics, thereby exerting varying influences on employment in China. Notably, the basic layer serves as the foundation of the AI industry, offering data and computing power support for AI applications. The technology layer represents the nucleus of the entire AI industry, encompassing various specialized technology domains that generate demand for skilled personnel. Lastly, the application layer constitutes the expansion of the AI industry, involving the exploration and integration of diverse AI technologies to address practical application scenarios (Agrawal et al., 2019; Wang et al., 2022; Xue et al., 2022; Altenried, 2020). Fig. 3 illustrates the specific structure of the AI industry.

In Fig. 3, the primary layer encompasses key components such as big data, cloud computing, and algorithms. Big data serves as the fundamental building block of AI, providing the necessary data for algorithm modeling. Within the primary layer, the position of a database development engineer holds significant importance, and the number of these positions reflects a growing trend in conjunction with AI's advancement. Cloud computing forms the technical foundation that enables independent learning for AI. By leveraging DL and other algorithms, cloud computing empowers AI with robust data processing capabilities. Consequently, there has been a substantial rise in employment opportunities in cloud computing roles, with enterprises like Alibaba Cloud offering comprehensive AI cloud services. Algorithms, as the continuous set of instructions for problem-solving and the methodological foundation of AI implementation, play a pivotal role in propelling AI's development. As a result, algorithm-related jobs have proliferated due to the increasing demand for AI.

The technical layer of the AI industry primarily encompasses speech recognition, natural language processing (NLP), and computer vision (CV). In the domain of speech recognition, machines employ various technical methods, including signal recognition and processing technology, to automatically recognize and convert human language into textual information. Speech recognition aims to enable computers to comprehend language in multimedia content. In contrast, NLP focuses on analyzing the intentions and meaning conveyed through language. Through the construction of machine algorithms and extensive language data inputs, NLP involves training language models and designing interactive application systems based on these models. CV leverages computer software and hardware to simulate the human visual system for pattern recognition. One prominent facet of CV is face image recognition, which finds wide-ranging applications in diverse sectors, including finance, security, and autonomous driving. As the scope of its applications continues to expand, face image recognition is poised to exert a consistently positive impact on the economy and employment landscape.

In the application layer, the primary objective is to facilitate the widespread adoption of AI technology. As the trend of “AI+” continues to gain momentum, the integration of AI applications with traditional industries becomes increasingly seamless. This integration is evident in sectors such as intelligent transportation, cultural tourism, government affairs, and social communication. Consequently, sizeable integrated technology companies specializing in AI technology have emerged,

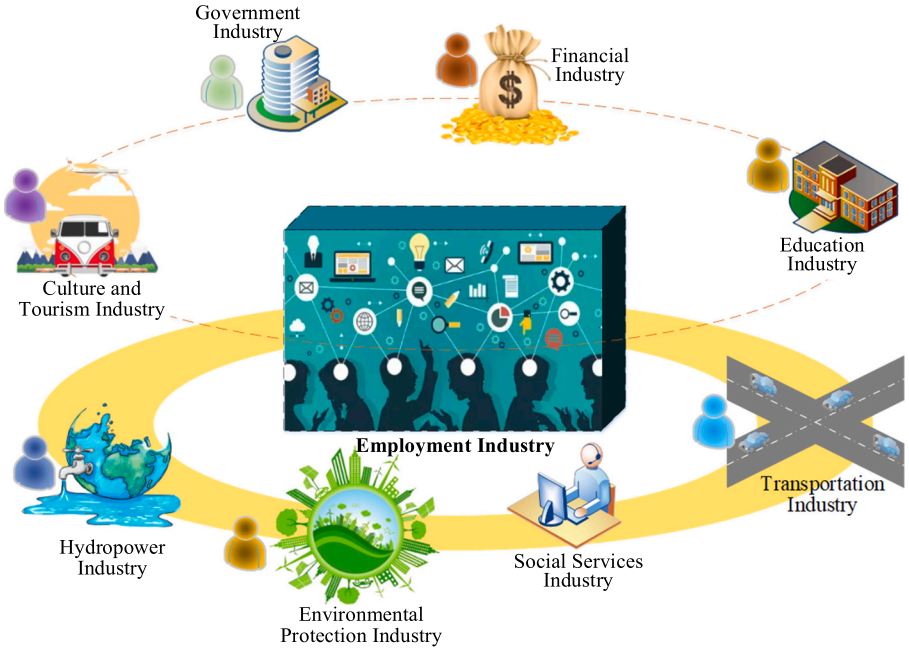

Fig. 2. Categories of industries in the labor force employment structure.

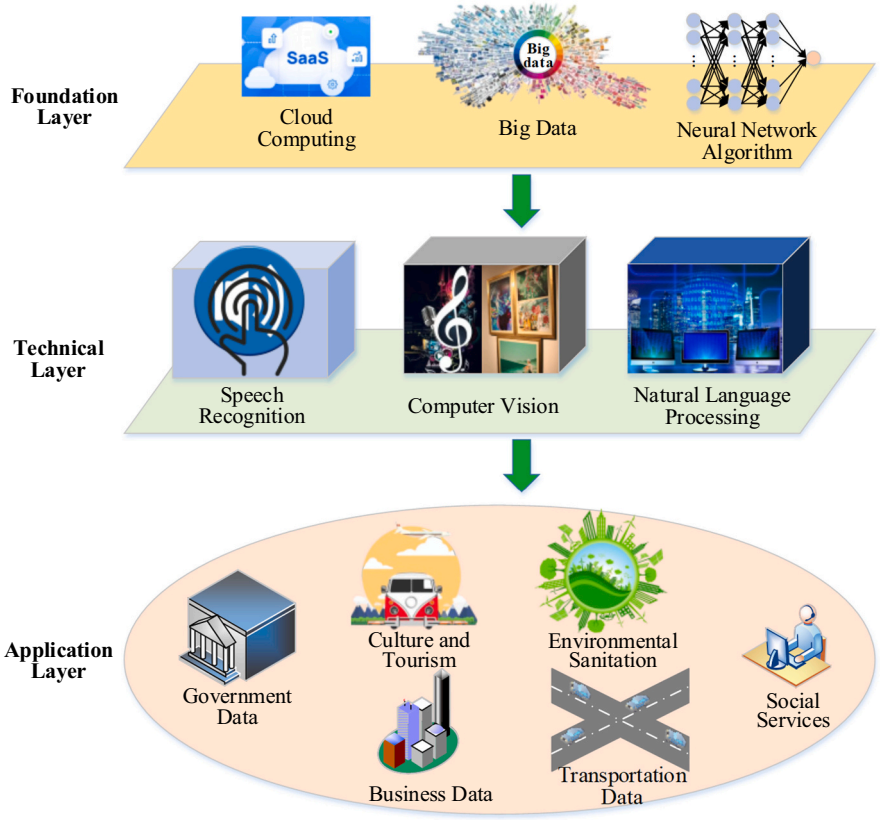

Fig. 3. Structure of the AI industry.

creating new employment opportunities and potentially increasing the demand for traditional jobs (Wang et al., 2022; Chen et al., 2023).

The employment situation of the labor force has been examined from the vantage points of the foundational, technical, and application layers. It is evident that all these levels generate novel employment opportunities and continuously influence the future economy and employment landscape. Nonetheless, further research is necessary to comprehend the impact on the structure of labor employment.

3.3. Mechanism and theoretical model construction analysis of labor employment and structural change in the AI industry

The primary objective is to understand the intricate influence of AI-related industries on the quantity and composition of labor employment. The analysis focuses on the distribution of employed individuals across different industries and the relative proportions of workers with varying skill levels. By employing the pervasive intelligence effect, the study delves into the underlying mechanisms through which AI impacts the structure of the labor force. The conceptual framework outlining the theoretical structure is illustrated in Fig. 4.

In Fig. 4, the AI industry undergoes a transformative process facilitated by key technologies such as big data, ANNs, and cloud computing. This transformation transcends the constraints of tacit knowledge and enables the integration of intelligent machines into both low- and high-skilled tasks. The analysis begins by considering a job market characterized by competition between two types of labor: high-skilled and low-skilled. High-skilled labor is defined by possessing a high school degree or higher, while low-skilled labor typically holds a technical secondary school education or below. The labor force does not directly generate products; instead, labor skills are applied to tasks along a continuous unit interval [0, 1], ultimately yielding a single final product denoted as Y. Each labor skill corresponds to a unique work task, establishing a one-to-one mapping relationship. Y represents the output of the final product, with its price normalized to 1. The production function of the final good Y is formulated as Eq. (1).

$$Y = exp\left[\int_0^1 lny(i)di\right] \tag{1}$$

Further calculations can be derived using Eq. (2).

$$lnY = \int_0^1 lny(i)di \tag{2}$$

In Eq. (2), $y(i)$ means the output of each task i. The production function of task i can be written as Eq. (3).

$$y(i) = A_L a_L(i) l(i) + A_H a_H(i) h(i) + A_K a_K(i) k(i) \tag{3}$$

In Eq. (3), A_L and A_H refer to the corresponding technical level of the high-skilled labor force and low-skilled labor force, respectively; A_K stands for the technical level of AI; $a_L(i)$, $a_H(i)$, and $a_K(i)$ represent the productivity of high-skilled, low-skilled, and AI workers, respectively; $l(i)$, $h(i)$, and $k(i)$ represent the number of tasks assigned to high-skilled, low-skilled, and AI workers. The production function Eq. (3) of task i means that low-skilled and high-skilled labor workers can perform and complete each task. There are comparative advantages of different skill execution and completion tasks.

Assuming that $a_L(i)/a_H(i)$ is a strictly decreasing function that is continuously differentiable, it can be inferred that the productivity of high-skilled labor is superior to that of low-skilled labor in more intricate tasks.

The distribution of production tasks across two skill levels is examined, taking into account the disparity in the assumed comparative advantages. This difference gives rise to a critical value, denoted as I, within the task interval of [0, 1]. In an equilibrium state, firms hire low-skilled labor for tasks where $i < I$, as the profit generated from performing these tasks exceeds that from employing high-skilled labor. Conversely, for tasks within the interval $i > I$, the profit from employing

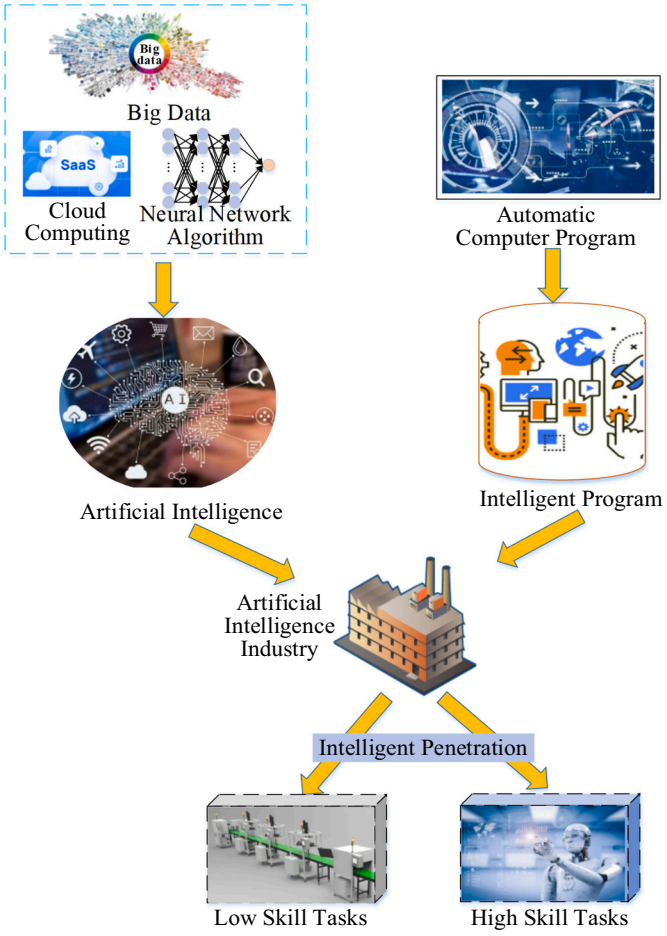

Fig. 4. Model frame of labor employment and structural change in the AI industry.

high-skilled labor surpasses that of low-skilled labor. Therefore, the following task allocation method can be derived: In equilibrium, there exists $0 < I < 1$. When $0 < i < I$, $h(i) = 0$; when $I < i < 1$, $l(i) = 0$.

In the equilibrium state, the completion of tasks within the intervals $[0, I]$ and $[I, 1]$ is solely attributed to low-skilled labor and high-skilled labor, respectively. These intervals, $[0, I]$ and $[I, 1]$, correspond to two distinct categories of tasks: routine tasks and non-routine cognitive tasks. Consequently, the production function Eq. (3) for task i can be effectively partitioned into Eqs. (4) and (5) to account for these different task categories.

$$y_L(i) = A_L a_L(i) l(i), 0 < i < I \tag{4}$$

$$y_H(i) = A_H a_H(i) h(i), I < i < 1 \tag{5}$$

Here, $p(i)$ represents the price of task i. In an equilibrium state, workers with the same skill level should receive the same wage denoted as w, as no worker would be willing to provide labor for a lower wage. The wage expression for each skilled worker can be formulated as follows:

$$w_L = p(i) A_L a_L(i), 0 < i < I \tag{6}$$

$$w_H = p(i) A_H a_H(i), I < i < 1 \tag{7}$$

Moreover, it is postulated that the marginal productivity of labor at the same skill level when performing different tasks is equivalent. This assumption implies that, for any given tasks i and i' within the interval [0,1], the following Eq. (8) is valid.

$$p(i) a_L(i) = p(i') a_L(i') = P_L \tag{8}$$

In addition, the price index (PL) represents the relative price of tasks executed by low-skilled workers. Similarly, for any given tasks i and i' within the interval [0, 1], the following expression is elucidated in Eq. (9).

$$p(i) a_H(i) = p(i') a_H(i') = P_H \tag{9}$$

P_H indicates the price index of tasks performed by low-skilled workers.

In equilibrium, the final product manufacturer endeavors to minimize costs for each task. By taking the first derivative of the cost minimization equation, the following expression can be derived for any given tasks i and i', as presented in Eq. (10).

$$p(i) y(i) = p(i') y(i') \tag{10}$$

The valuation of the final product is denoted as $P = 1$. By considering this valuation, the following Eq. (11) can be derived.

$$p(i) y(i) = Y, i \in [0, 1] \tag{11}$$

Considering the endogeneity of the critical value I, it is crucial to incorporate the determination of this value into the analysis of the model. When examining two tasks, denoted as i and i', performed by low-skilled workers, it becomes evident that the output Y remains constant. This observation gives rise to Eq. (12).

$$p(i) A_L a_L(i) l(i) = p(i') A_L a_L(i') l(i') \tag{12}$$

Eq. (13) can be derived by incorporating Eq. (8) introduced earlier.

$$A_L P_L l(i) = A_L P_L l(i') \tag{13}$$

Thus, for any values of $0 < i$ and $i' < 1$, the labor demand can be represented as $l(i) = l(i')$.

Similarly, for any values of $I < i$ and $i' < 1$, the labor demand can be represented as $h(i) = h(i')$.

The task allocation analysis following the integration of AI technology is further examined, as depicted in Fig. 5. Assuming the presence of task intervals $[I_{a1}, I_{b1}] \subset [0, I_c]$ and $[I_{a2}, I_{b2}] \subset [0, I_c]$, this study postulates the manifestation of AI techniques' pervasive intelligence in low-skill tasks and high-skill tasks, respectively. When AI capital $a_K(i)$ is sufficiently invested in $[I_{a1}, I_{b1}]$ and $[I_{a2}, I_{b2}]$, it becomes more cost-effective for AI to undertake tasks within the intervals $[I_{a1}, I_{b1}]$ and $[I_{a2}, I_{b2}]$. For tasks outside the intervals $[I_{a1}, I_{b1}]$ and $[I_{a2}, I_{b2}]$, the assumption is made that $a_K(i) = 0$. To simplify the analysis, the interval lengths are defined as $\varepsilon_1 = I_{b1} - I_{a1}$ and $\varepsilon_2 = I_{b2} - I_{a2}$.

After the introduction of AI, a new critical value I_c emerges, where $0 < I_{a1} < I_{b1} < I_{a1} < I_{a2} < I_{b2} < 1$. Tasks falling within the intervals $[I_{a1}, I_{b1}]$ and $[I_{a2}, I_{b2}]$ are now handled by AI. In this context, the effects of changes in the AI input intervals $[I_{a1}, I_{b1}]$ and $[I_{a2}, I_{b2}]$ on the critical value I_c are analyzed. Furthermore, the impact of variations in the critical values I_a and I_b on the size of the new task intervals $[0, I_{a1}] \cup [I_{b1}, I_c]$ and $[I_c, I_{a2}] \cup [I_{b2}, 1]$ is investigated. These analyses aim to shed light on the interplay between task interval transformations and the resulting

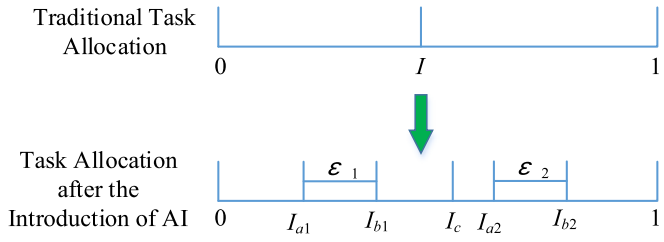

Fig. 5. Changes in task allocation from traditional task allocation to the introduction of AI technology.

labor demand in relevant industries following the adoption of AI technology.

The analysis examines the influence of AI technology on the income disparity among employed workers. Capital income is utilized to standardize the wage income of both types of labor, namely w_L and w_H. The expressions for capital income and wage income are as follows:

$$w_L = \frac{W_L}{Rk(i)} \tag{14}$$

$$w_H = \frac{W_H}{Rk(i)} \tag{15}$$

Here, R, w_L, and w_H represent the rent of machines and the wage of low-skilled and high-skilled labor, respectively.

Then, the labor skill premium is calculated according to:

$$ln(w_H/w_L) = ln(W_H/Rk(i)) - ln(W_L/Rk(i)) \tag{16}$$

Eq. (17) can be obtained by further arranging Eq. (16).

$$dln(w_H/w_L) = dlnw_H - dlnw_L \tag{17}$$

The impact of intelligence penetration on the income disparity between different skill levels is determined by the combined effect of intelligence penetration I_1 for low-end tasks and intelligence penetration I_2 for high-end tasks. The influence of pervasive intelligence on the income level of high-skilled labor can be observed through the following equations:

$$dlnw_H = \theta_{21}dI_1 + (\theta_{22} + \theta_{23})dI_2 \tag{18}$$

$$\theta_{21} = \theta_{22} = -\frac{1}{\sigma + \varepsilon_2}(1 - I_c)^{-1} \tag{19}$$

$$\theta_{23} = -\frac{1}{\sigma + \varepsilon_2}\frac{a_H(I_2)^{\sigma-1}}{\int a_H(I_2)^{\sigma-1}dI_2} \tag{20}$$

Here, σ refers to the elasticity of factor substitution.

Further analysis is conducted to examine the influence of on the income level of low-skilled labor, as presented in Eq. (21).

$$dlnw_L = (\theta_{11} + \theta_{12})dI_1 + \theta_{13}dI_2 \tag{21}$$

$$\theta_{12} = \theta_{13} = -\frac{1}{\sigma + \varepsilon_1}I_c^{-1} \tag{22}$$

$$\theta_{11} = -\frac{1}{\sigma + \varepsilon_1}\frac{a_L(I_1)^{\sigma-1}}{\int a_L(I_1)^{\sigma-1}dI_1} \tag{23}$$

Based on Eqs. (18), (21), and (17), the variability can be summarized as Eq. (24).

$$\begin{aligned} &dln(w_H/w_L) \\ &= (\theta_{21} - \theta_{11} - \theta_{12})dI_1 + (\theta_{22} + \theta_{23} - \theta_{13})dI_2 \end{aligned} \tag{24}$$

Higher levels of education are associated with higher working time elasticity and labor participation elasticity in the labor force (Schiller and McMahon, 2019). As $\varepsilon_2 > \varepsilon_1$, it implies that $\frac{1}{\sigma+\varepsilon_1} > \frac{1}{\sigma+\varepsilon_2}$. Considering that the promotion of pervasive intelligence I_1 for low-end work tasks, $dI_1 > dI_2$, is more challenging than promoting pervasive intelligence I_2 for high-end work tasks. Finally, it can be observed that the first and second items in Eq. (24) are positive, indicating that the income gap between high and low-skilled laborers may widen after the penetration of AI in relevant industries. The causal cycle diagram of system dynamics, illustrating the intricate relationship between system dynamics and market economic structure, is presented in Fig. 6.

The causal loop model within the domain of system dynamics, as depicted in Fig. 6, offers valuable insights into the factors and mechanisms driving changes in labor force structure. Utilizing this model allows for a comprehensive examination of the interdependencies among

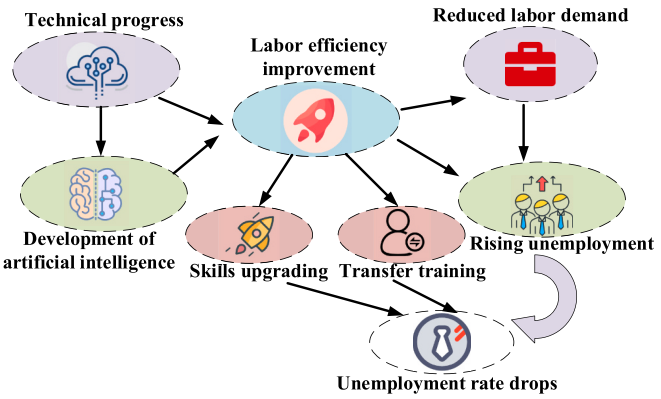

Fig. 6. System dynamics of the impact of AI on employment in the digital economy.

different variables and a quantitative assessment of their impacts. Such an approach enables a deeper understanding of the influence of the AI industry on labor employment quantity and structure. Through the analysis of this model, discerning the long-term implications of labor force structure changes and devising strategies to foster digital economic development becomes possible. Moreover, it facilitates the identification of effective measures for long-term prediction and intervention in the development of AI and the digital economy industry.

3.4. Technical framework analysis of labor economics and system dynamics

Labor economics is intricately intertwined with the development and transformation of society. Firstly, it is closely linked to the emergence and progression of industrialization. The size and quality of the labor force have emerged as pivotal factors propelling economic advancement during this process. Secondly, labor economics is intricately connected to technological progress and innovation, which have ushered in new prospects and challenges within the labor market. The rise of novel industries, the decline of traditional ones, the proliferation of diverse occupations, and the specialization of skills all exemplify the impact of technological advancements on the labor market. The field of labor economics endeavors to study and analyze these shifts, playing a crucial role in understanding labor adaptability and market demands. Thirdly, modernization and economic globalization exert a profound impact on labor economics. Global labor mobility facilitates the optimal allocation of human resources and the reduction of labor costs. Nonetheless, it also entails the risk of unemployment and introduces uncertain social trends. To effectively navigate these changes, it becomes imperative to develop targeted theories and methodologies within the context of globalization and modernization. These actions enable us to better respond to societal challenges and transformations. Furthermore, the labor market can be comprehensively analyzed through system dynamics modeling, thereby illuminating feedback mechanisms and dynamic equilibrium. For instance, when the employment rate declines, the unemployment rate inevitably rises, prompting an increase in job seekers and subsequently impacting both employment and unemployment rates. By simulating these feedback mechanisms, a deeper understanding of dynamic equilibrium and stability in the labor market can be achieved. Consequently, this understanding can inform the development of effective labor policies and management strategies, providing valuable decision support. The integration of system dynamics modeling into labor economics research provides a valuable framework for analyzing the intricate dynamics within the labor market. This approach enables a comprehensive examination of the interplay and feedback loops among diverse factors that shape employment and market structure. By considering the interdependencies between the AI industry, labor employment, and market dynamics, this study can profoundly understand their long-term

implications and identify sustainable development pathways. The application of system dynamics models allows for an assessment of the labor market's sustainability in the digital economy era, encompassing crucial aspects such as labor education, skill development, and potential job displacement risks. This holistic approach facilitates an analysis of the current state of labor employment. It offers predictions and recommendations for future policy-making, aiming to foster sustainable and inclusive labor market development within the AI industry and the digital economic environment.

3.5. The empirical analysis

The study hypothesis under investigation is: How will the future employment structure of the Chinese labor force be influenced by the development of the digital economy, AI, and policy changes?

To address this research hypothesis, simulations can be conducted with varying parameter values to analyze the impact of key variables, including the proportion of the digital economy, the adoption rate of digital technologies, the level of policy support, and the level of AI technology, on the future employment structure. In addition, conducting various sensitivity analyses, such as the one mentioned in the baseline study, can validate the robustness and reliability of the research findings.

Moreover, employing an agent-based model approach allows for the categorization of variables into three types: policy variables, digital economy variables, and AI technology variables. The following are the specific simulation specifications and parameter settings:

Policy variables: Policy variables can be represented as binary variables, indicating whether the country promotes skill training and employment based on talent development. In this context, '1' signifies the implementation of the policy, while '0' represents the absence of policy implementation.

Digital economy variables: Digital economy variables can be represented as binary variables, reflecting the adoption rate of digital technologies and the proportion of the digital economy. Here, '1' denotes a state of economic growth, while '0' signifies a state of economic decline.

AI technology variables: AI technology variables can be represented as binary variables, indicating the level of AI technology development. In this case, '1' denotes a state of advanced AI technology, while '0' signifies a state of less developed AI technology.

The measurement model used here follows the approach proposed by Hamstra et al. (2019) (Hamstra et al., 2019) for reference. This measurement model aims to analyze the changes in the labor force structure under the influence of pervasive intelligence, as outlined in the constructed model of labor force employment and structural transformation within the AI industry.

$$\frac{Low_{it}}{High_{it}} = \alpha_1 + \alpha_2 lnAI_{it} + \alpha_3 X_{it} + \lambda_i + \nu_t + \xi_{it} \tag{25}$$

In Eq. (25), i represents province; t refers to time; α_1, α_2, and α_3 refer to the constant term; λ_i indicates an individual factor; ν_t stands for the time factor; ξ_{it} expresses the random error term.

The dependent variable $\frac{Low_{it}}{High_{it}}$ represents the proportion of the low-skilled labor force to the high-skilled labor force in each province and city during period t. The key independent variable "AI" refers to the presence of AI, while "X_{it}" denotes a set of control variables affecting both the high-skilled and low-skilled labor forces. These control variables include the trade outsourcing index (OS), industrial structure upgrading (FIR), urbanization level (Ur), and investment intensity in research and development (R&D). For this empirical analysis, a panel dataset covering 30 provinces, autonomous regions, and municipalities directly under the Central Government, spanning the years 2006 to 2020 (excluding Tibet), is selected. The definitions and detailed descriptions of each variable are provided in Table 1.

Table 1 presents the empirical analysis conducted here on the employment quantity and structure of China's labor force during the period 2006 to 2020. The descriptive statistics of the data are provided below. The study utilizes a sample data size of 400, sourced from the China Statistical Yearbook (CSY) and National Research Network Database. The variables used in the analysis are denoted as follows: $\frac{Low_{it}}{High_{it}}$ represents the proportion of the low-skill labor force to the high-skill labor force; "FIR" refers to industrial structure upgrading; "AI" indicates the number of jobs in the AI industry; "OS" represents the trade outsourcing index; "Ur" signifies the level of urbanization; "R&D" denotes investment intensity in research and development. The data extraction process begins with accessing the CSY, followed by data preprocessing. Subsequently, the obtained employment industry structure data is input into the structural change model for further analysis. In terms of the sample characteristics, this study utilizes empirical analysis data on the employment quantity and structure of the Chinese labor force spanning the period from 2006 to 2020. The sample comprises 400 observations, and data collection is sourced from reputable outlets such as the China Statistical Yearbook and the National Research Network Database. A multiple regression model is employed to assess and model the data for the analysis. Statistical software programs, including SPSS, STATA, and R, are utilized for calculations and data analysis.

To further investigate and validate the relationship between the independent and dependent variables in the model, additional analyses such as covariance analysis, F-test, and t-test are conducted. Conclusions are drawn based on the statistical results. The following methods are employed to ensure the reliability and credibility of the research findings:

Firstly, two different approaches are utilized to preprocess and model the data to verify the consistency and stability of the results. Secondly, sensitivity analysis was performed to identify factors that may influence the research outcomes. Repeat experiments are conducted to test the robustness and reliability of the research conclusions. The sample reliability is found to be 86.7 %, while the sample stability is determined to be 87.9 %. These results affirm that the research findings successfully pass the robustness test.

4. Results and discussion

4.1. Current situation of the number and structure of labor force employed in China

This section aims to gain a comprehensive understanding of the current state of labor employment in China, specifically focusing on the number and structure of labor employment. Therefore, an analysis of the

Table 1
Types of empirical variables and sample statistics.

	Variable name	Variable symbol	Sample size	Data source
Explained variable	Low-skill share/ high skillshare	$\frac{Low_{it}}{High_{it}}$	400	China Statistical Yearbook
Explanatory variables	AI	AI	400	China Statistical Yearbook
Control variables	Trade outsourcing index	OS	400	National research network database
	Upgrading of industrial structure	FIR	400	China Statistical Yearbook
	Level of urbanization	Ur	400	China Statistical Yearbook
	Investment intensity	R&D	400	China Statistical Yearbook

education level structure and average wage income level is conducted, as depicted in Figs. 7 and 8, respectively.

According to Fig. 7, it can be observed that from 2006 to 2020, the proportion of the labor force with junior high school education and below exhibited a consistent downward trend. Conversely, the proportion of the labor force with technical secondary school, senior high school, junior college, and bachelor's degree or above demonstrated a continuous increase over the years. Furthermore, the educational level of the labor force in the primary industry, secondary industry, and tertiary industry is examined individually. In the primary and secondary industries, there was a significant decrease in the proportion of the labor force with primary school education. In contrast, the proportion of the labor force with other educational levels experienced a steady but modest increase. In the tertiary industry, the proportion of the labor force with junior high school education and below exhibited a substantial decline year by year. On the contrary, the proportion of the labor force with technical secondary school, senior high school, junior college, and bachelor's degree or above displayed a significant yearly increase. Thus, the analysis of the employment structure based on educational levels indicates an overall rise in the educational attainment of the labor force, reflecting a trend of upward mobility in employment.

Fig. 8 presents an analysis of the proportion trend of average wage income levels among different labor forces in the total number of employees. The findings indicate that, overall, the proportion of individuals earning below 5000 RMB is gradually decreasing, yet it still exceeds 60 %. In comparison, the proportion of the labor force with an average wage income level between 5000–10,000 RMB and above 10,000 RMB remains relatively small but shows an increasing trend year by year. Furthermore, the average wage income level analysis reveals that, as the year progresses, the average wage income in the primary, secondary, and tertiary industries falls below 5000 RMB. However, there is a gradual increase in the proportion of the labor force with an average income level ranging from 5000 to 10,000 RMB and above 10,000 RMB. Among the three industries, the primary industry exhibits the highest wage income level, followed by the secondary industry, while the tertiary industry has the lowest average wage income level. Based on the benchmark salary of 5000 RMB, the employment distribution in China's national economy demonstrates a polarization in industry employment.

4.2. Analysis of empirical results of the model

In the empirical analysis of the model, a unit root test is conducted initially to prevent spurious regression. To ensure robustness, three widely used unit root test methods, namely the Levin-Lin-Chu (LLC) test (Rudko et al., 2021), the Im-Pesaran-Shin (IPS) test (Li, 2022), and the Breitung test (Popelo et al., 2021), are jointly employed in this study. This approach helps mitigate potential errors stemming from method selection. Stationarity of the data is confirmed when the null hypothesis of the presence of a unit root is rejected by at least two of the tests. Considering that all economic and control variables exhibit time trends, the unit root test is performed on a model incorporating these trends. The results of the unit root test are presented in Table 2 and visualized in Fig. 9.

Table 2 and Fig. 9 showcase the application of three widely used unit root test methods, namely the LLC test, IPS test, and Breitung test, to examine the empirical data. These tests are conducted on the explained variables, core explanatory variables, and control variables in this study. The results indicate that all variables are stationary, as confirmed by the LLC, IPS, and Breitung tests. Thus, the variables constructed for the empirical analysis in this study are deemed valid.

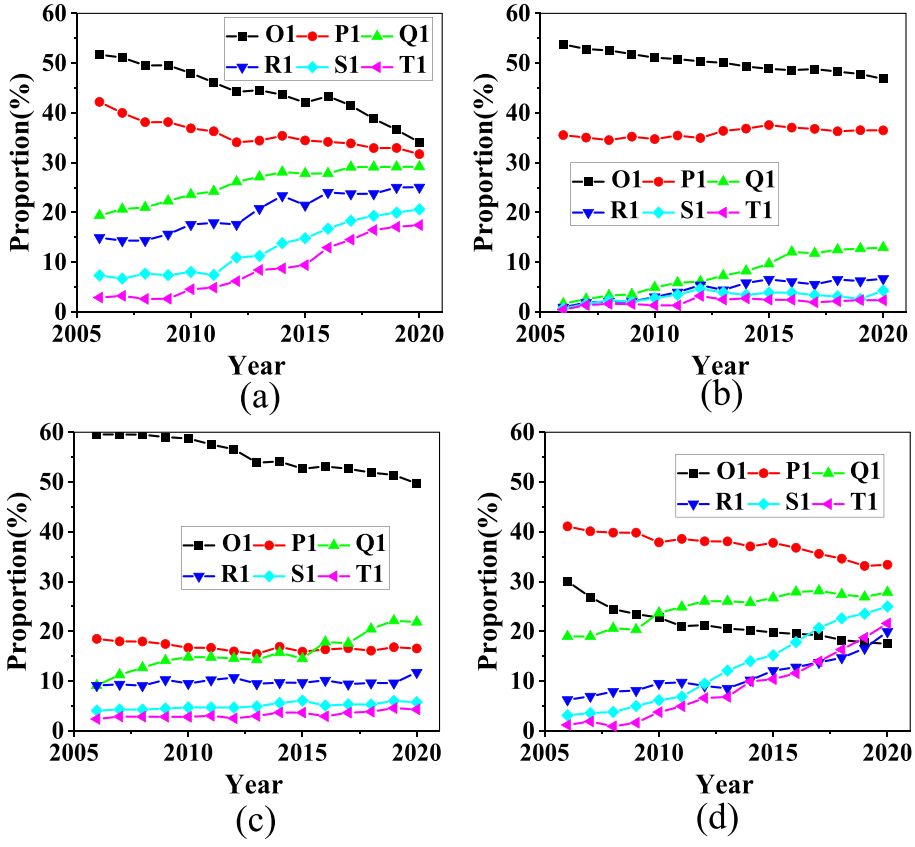

Fig. 7. Changes in the structural proportion of educational attainment of the labor force from 2006 to 2020 (a. changes in the educational structure of the national labor force; b. changes in the primary industry's educational structure of the labor force; c. changes in the education structure of the labor force in the secondary industry; d. changes in the education structure of the labor force in the tertiary industry) (Here, O1: primary school; P1: junior high school; Q1 technical secondary school; R1: high school; R1 college; S1 Bachelor degree or above).

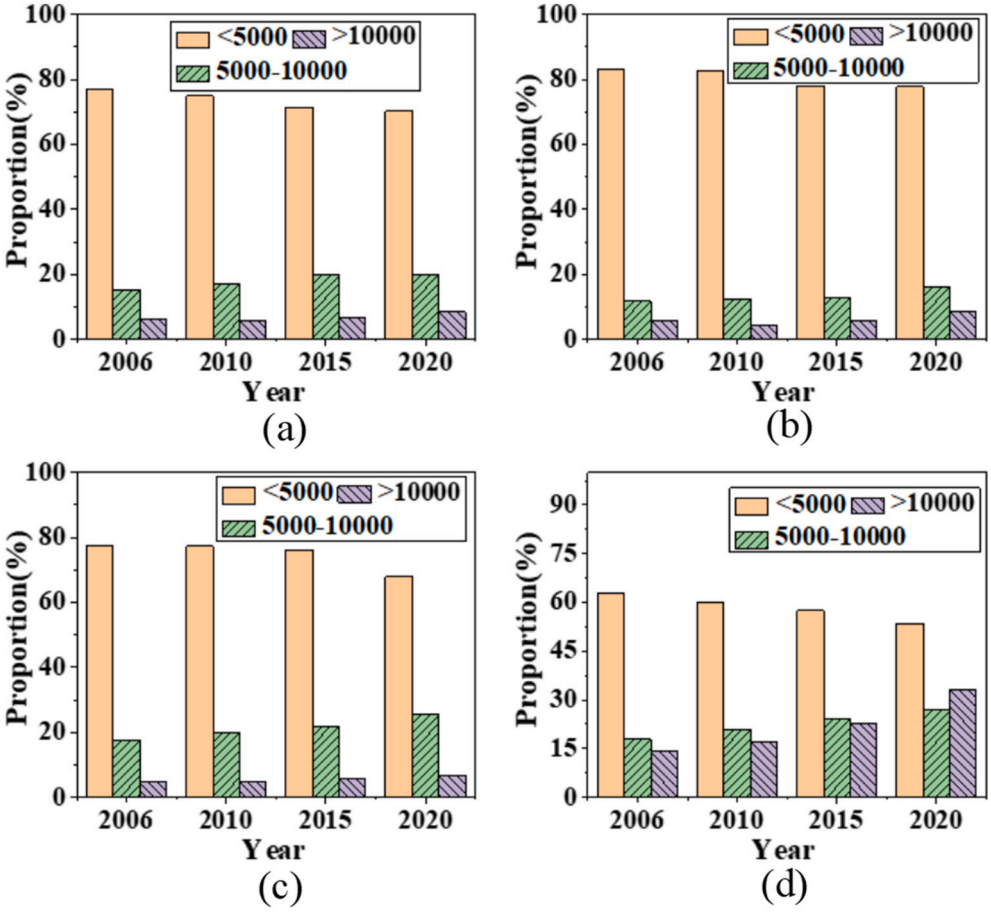

Fig. 8. Variations in the structural share of the average wage income level of the labor force during 2006–2020 (a. changes in the structure of national average wage income; b. structural changes of the average wage income level of the labor force in the primary industry; c. structural changes of the average wage income level of labor in the secondary industry; d. structural changes of the average wage income level of the tertiary industry labor force).

Table 2
Empirical test results of unit root method.

Variable	$\frac{Low_{it}}{High_{it}}$	AI	OS	FIR	Ur	R&D
Whether smooth	Smooth	Smooth	Smooth	Smooth	Smooth	Smooth

Subsequently, a benchmark regression analysis is performed on each variable under the primary, secondary, and tertiary industries, employing fixed effect, random effect, and mixed effect models. The outcomes of these analyses are presented in Table 3 and Fig. 10.

Table 3 and Fig. 10 demonstrate that AI has a lag effect on the real labor market, indicating that the specific application of AI follows an investment period. Consequently, the AI variables are included in the model estimation with a lag period.

Based on the results of the benchmark regression analysis, it is evident that AI significantly contributes to the intelligent enhancement of both low-skilled and high-skilled labor in China's primary, secondary, and tertiary industries. Moreover, the impact of industrial structure on the demand for low-skilled labor outweighs its impact on high-skilled labor following intelligent upgrading. Thus, the assumption that AI technology penetration in relevant fields encompasses both pervasive intelligence in low-end tasks and pervasive intelligence in high-end tasks aligns with the findings of the baseline recovery analysis.

The robustness of each variable is rigorously examined, and the corresponding results are presented in Table 4 and Fig. 11.

The robustness test incorporates the lag term of the explained variable into the baseline equation and employs the dynamic panel model to examine the impact of AI on labor employment structure. Introducing the lag term of the explained variable helps address the

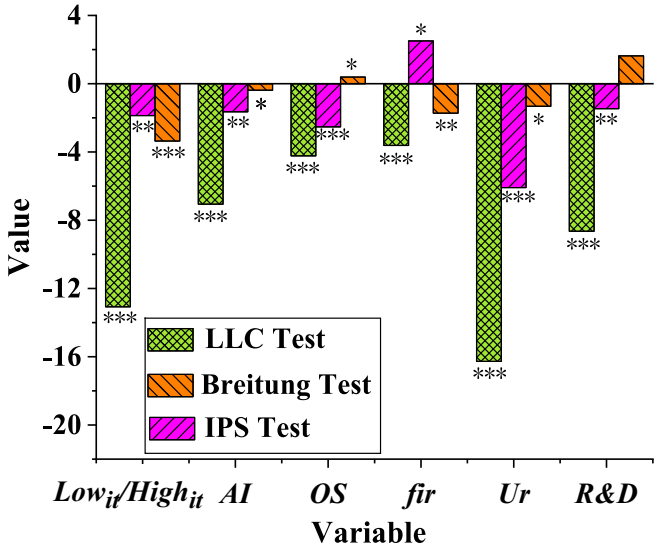

Fig. 9. Empirical test results of the unit root method.
(Note: ***, **, and * respectively indicate rejection of the null hypothesis of the existence of unit root at the significance level of 1 %, 5 %, and 10 %).

Table 3
Regression estimation results of labor force structure in AI industry.

Variable	$\frac{Low_{it}}{High_{it}}$		
	The first industry	The second industry	The third industry
Constant term	17.52***	8.62***	1.54***
	(1.53)	(1.33)	(0.21)
Individual factors	Yes	Yes	Yes
Time factor	Yes	Yes	Yes
Hausman test	Fixed effects	Fixed effects	Fixed effects
N	400	400	400
R2	0.83	0.82	0.80

Note: Standard errors are in parentheses. *, **, and *** indicate passing the test at the significance level of 10 %, 5 %, and 1 %, respectively.

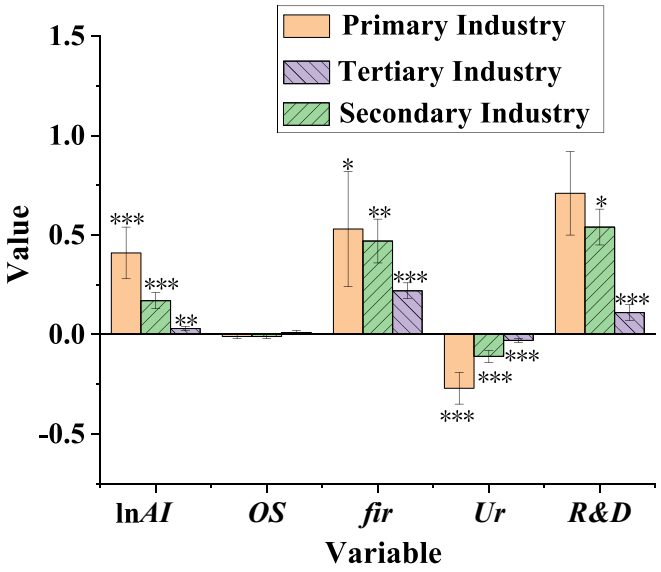

Fig. 10. Variables estimated by regression of labor force structure in the AI industry.
(Note: *, **, and *** respectively pass the test at the significance level of 10 %, 5 %, and 1 %).

Table 4
Robustness test results of labor force structure in AI industry.

	Dynamic panel estimation method		
	The first industry	The second industry	The third industry
Constant term	14.53***	8.77***	−0.31***
	(0.53)	(0.33)	(0.21)
$\left(\frac{Low}{High}\right)_{it-1}$	0.47***	0.31***	0.59***
	(0.02)	(0.01)	(0.02)
N	400	400	400
Sargan test P value	1.00	1.00	1.00
First-order serial correlation test (AR (1) test)	0.01	0.01	0.00
Second serial correlation test (AR (2) test)	0.75	0.68	0.61

Note: Standard errors are in parentheses. *, **, and *** indicate passing the test at the significance level of 10 %, 5 %, and 1 %, respectively.

potential endogeneity issue. The First Difference Generalized Method of Moments is employed for the robustness test to ensure robustness. This model undergoes two serial correlation tests, namely the first-order serial correlation test and the second-order serial correlation test. The validity of the model relies on the acceptance of the null hypothesis by the second-order serial correlation test, indicating the absence of first-order difference and autocorrelation in the perturbation term of the difference model. The robustness test confirms the validity of the model as all tests are successfully passed. The significance and direction of the coefficient for AI remain consistent with the benchmark regression results. This finding supports the model assumption that the penetration of AI into relevant industries holds the potential to widen the income gap between the high-skilled and low-skilled labor force.

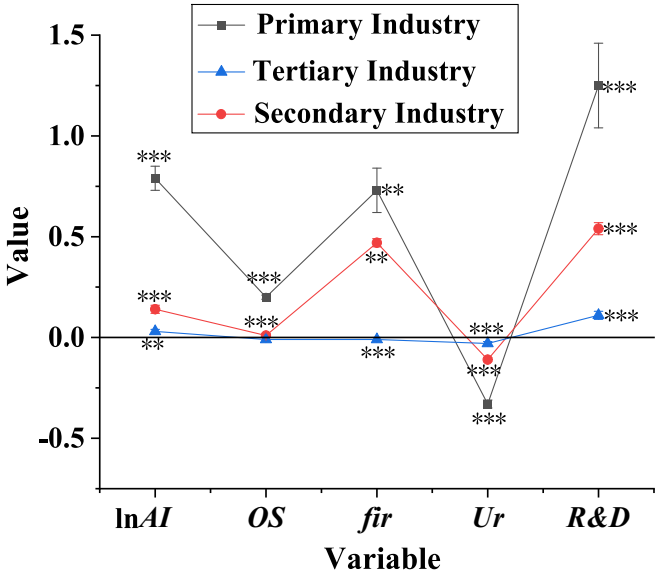

Fig. 11. Robustness test results of each variable (Note: *, ** and *** respectively indicate that they pass the test at the significance level of 10 %, 5 %, and 1 %).

5. Discussion

This study aims to analyze the current status of labor employment in China, focusing on the education level and average income level structure (Ma et al., 2022). The findings reveal an overall increasing trend in labor education, accompanied by employment upgrading and employment polarization within China's national economy industry. Empirical analysis is conducted to validate the constructed model, drawing insights from previous studies (Nam et al., 2021; Chen et al., 2023; Damioli et al., 2021). The results confirm the hypothesis that AI technology permeates various industries, encompassing both low-end and high-end tasks through pervasive intelligence. This hypothesis is further supported by robustness analysis, which demonstrates that the penetration of AI in related industries expands and widens the income gap between high-skilled and low-skilled labor (Kurt, 2019).

The research findings indicate a discernible trend in the Chinese labor force, characterized by a general elevation in educational attainment and enhanced employment prospects. Nonetheless, a conspicuous polarization in industrial employment is also observable. An analysis of China's policies related to digital economic development and the advancements in digital technologies underscores the constructive influence of the digital economy on China's broader economic growth. Furthermore, it facilitates the seamless advancement of digital transformation.

The principal contribution of this study lies in its investigation of the impact of the AI industry on the employment structure of the Chinese labor force, underpinned by the framework of Marxist theory. This study adopts a quantitative and structural approach, integrating technological methodologies to forecast transformational patterns, analyze their societal ramifications, and offer actionable recommendations. Consequently, the research findings underscore the substantial significance of the AI industry's influence on the labor force's quantity and composition,

imparting valuable insights for policymakers and stakeholders alike.

The alignment of these research findings with existing literature underscores the global significance of the AI industry's influence on the labor market and its consequential effects. The results highlight how AI technology can be leveraged to achieve a high degree of employment digitalization and successfully transform the structure of labor force employment. Ultimately, this study offers policymakers and stakeholders crucial insights to address the challenges posed by the labor market in the digital era, enhance national competitiveness, and promote sustainable development.

The concurrence of these research findings with the existing literature underscores the global relevance of the AI industry's impact on the labor market and its consequential implications. The results highlight the potential of AI technology to be harnessed for achieving a high level of employment digitalization and effectively reshaping the structure of labor force employment. Ultimately, this study furnishes policymakers and stakeholders with invaluable insights for addressing the challenges posed by the labor market in the digital era, enhancing national competitiveness, and promoting sustainable development.

6. Conclusion

This study is grounded in examining the actual development and application level of AI in China, coupled with an analysis of the current labor force structure. By scrutinizing the educational level and average income level structure, it is observed that the educational level of the labor force is on an upward trajectory. Through in-depth analyses of the AI industry's structure, theoretical research on the digital economy, the mechanism by which the digital economy contributes to economic growth, and panel data analysis on the impact of the digital economy on regional economic growth, it becomes evident that each region must identify the appropriate positioning for digital economy development. The Internet has added impetus to economic progress, fostering new industrial models and structures. Digital sectors like AI, cloud computing, and big data necessitate a significant influx of digital professionals and capital investment. Empirical findings demonstrate that the AI industry contributes to both the “upgrading” and “polarization” of high-skilled and low-skilled labor, ultimately driving favorable overall economic development trends. Based on these results, the study proposes the concept of cultivating versatile, high-skilled talents. Three measures are suggested: fostering vocational education and training for acquiring new technologies, promoting the application of AI technology in the real economy, and offering a reference framework for the AI industry's subsequent impact on the digital economy's labor market. However, it is essential to acknowledge that this study has some limitations, such as the absence of an analysis of the macroeconomic model when constructing the labor employment structure model based on AI. Future research should delve into the evolution path of the labor force employment structure, taking into account the impact of AI technology innovation and the dynamic interplay with external factors, such as the educational level of the labor force.

CRediT authorship contribution statement

Zhuo Zhang contributed to conception and design of the study. Zhuo Zhang wrote the manuscript. The authors contributed to manuscript revision, read, and approved the submitted version.

Declaration of competing interest

The author has no conflict of interest.

Data availability

The raw data supporting the conclusions of this article will be made available by the authors, without undue reservation.

Appendix A

The relationship between the economic model equation adopted in the study and the empirical data can be elucidated as follows: The economic production model for commodity Y is established by linking products in the labor market with labor production skills, as depicted in Eqs. (1) and (2):

$$Y = exp\left[\int_0^1 lny(i)di\right] \tag{1}$$

$$lnY = \int_0^1 lny(i)di \tag{2}$$

i refers to the working time of workers. For each production task $y(i)$, the ultimate labor product is denoted as Y, and the task output of labor production is calculated via Eq. (3).

$$y(i) = A_L a_L(i) l(i) + A_H a_H(i) h(i) + A_K a_K(i) k(i) \tag{3}$$

In Eq. (3), A_L and A_H refer to the corresponding technical level of high and low-skilled labor force; A_K means the technical level of AI industry; $a_L(i)$ $a_H(i)$, and $a_K(i)$ stand for the productivity of high-skilled, low-skilled, AI workers respectively; $l(i)$, $h(i)$, $k(i)$ represent the number of tasks assigned to high-skilled, low-skilled, and AI workers. The task output of high and low skilled labors is calculated respectively, and the expression can be written as Eqs. (4) and (5):

$$y_L(i) = A_L a_L(i) l(i), 0 < i < I \tag{4}$$

$$y_H(i) = A_H a_H(i) h(i), I < i < 1 \tag{5}$$

Eqs. (6) and (7) describe the labor price and wages produced by workers.

$$w_L = p(i) A_L a_L(i), 0 < i < I \tag{6}$$

$$w_H = p(i) A_H a_H(i), I < i < 1 \tag{7}$$

$\boldsymbol{w_L}$ and $\boldsymbol{w_H}$ represent the wage level of high and low-skilled labor force. The marginal productivity of labor with the same skill level is equal when performing different tasks.

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